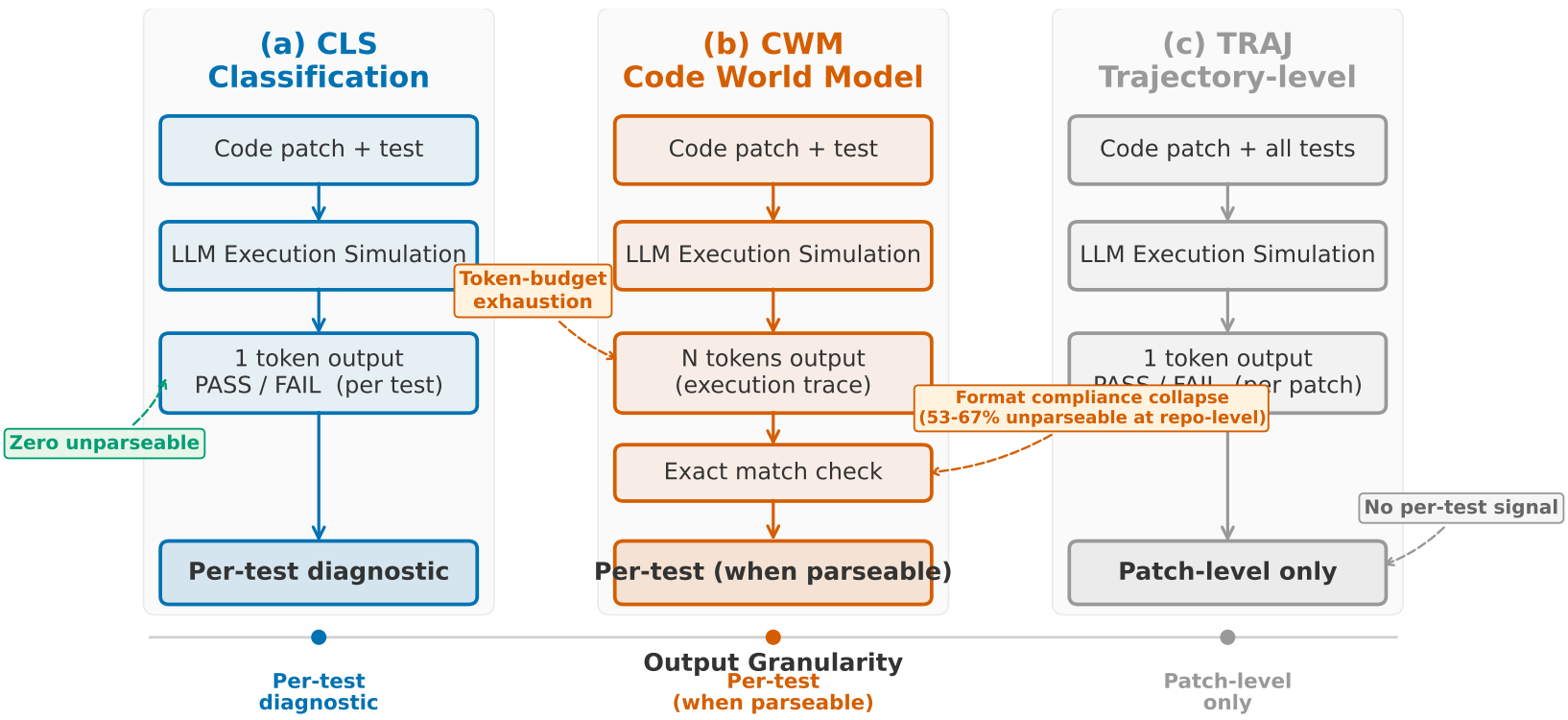
(a) CLS
Classification
Code patch + test
LLM Execution Simulation
1 token output
PASS / FAIL (per test)
Per-test diagnostic
Zero unparseable
(b) CWM
Code World Model
Code patch + test
LLM Execution Simulation
N tokens output
(execution trace)
Exact match check
Per-test (when parseable)
Token-budget exhaustion
(c) TRAJ
Trajectory-level
Code patch + all tests
LLM Execution Simulation
1 token output
PASS / FAIL (per patch)
Patch-level only
Format compliance collapse
(53-67% unparseable at repo-level)
No per-test signal
Output Granularity
Per-test diagnostic
Per-test (when parseable)
Patch-level only